Earth's Interior, Rocks & Plate Tectonics

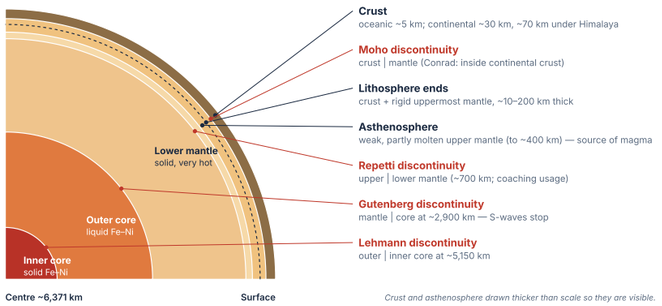
GS-I: salient features of world physical geography; geophysical phenomena — earthquakes, tsunami, volcanic activity · Prelims: physical geography of India and the world

7 Mains PYQs 2013–2025
Prelims ~1 a year since 2023, always a narrow fact · Tier B, Mains-weighted · draw a diagram in every answer

THE ANSWER SPINE 1 Interior & its layers → 2 Rocks → 3 Drift → spreading → plates → 4 Boundaries & belts → 5 Island arcs → 6 Plumes & the Wilson cycle

① INTERIOR · HOW WE KNOW · LAYERS

Prelims 2009, 2023



Crust and asthenosphere drawn thicker than scale so they are visible.

- **Direct:** drilling (Kola, ~12 km), volcanic material. **Indirect:** seismic waves (main), density/pressure/temperature, meteorites, gravity, magnetism.
- **Crust** ~5 km oceanic (basaltic, sima) / ~30 km continental (granitic, sial), ~70 km under the Himalaya · **Mantle** to ~2,900 km · **outer core** liquid Fe–Ni · **inner core** solid.
- **Lithosphere** (rigid, 10–200 km) rides on the **asthenosphere** (weak, to ~400 km, source of magma).

P = longitudinal, all states, first · **S** = transverse, solids only · surface waves last, most destructive. S absent beyond 105° → **liquid outer core**; P gap 105°–145°.

② ROCKS

GS-I 2022 · 10M

Igneous (primary)	Sedimentary	Metamorphic
Magma cools. Intrusive granite (coarse) · extrusive basalt (fine). No fossils, no layers. **Deccan Traps**	Clastic sandstone/shale · organic limestone/coal · chemical salt. Layers, fossils. **Vindhyan, Gondwana coal**	Heat + pressure. Limestone→marble · sandstone→quartzite · shale→slate · granite→gneiss. **Makrana**

Clay is **porous but impermeable** (Prelims 2025) · Vosges = **block** mountain (Prelims 2024) · quartzite ← sandstone.

Rock cycle: magma → igneous → (weathering, deposition) → sedimentary → (heat, pressure) → metamorphic → (melting) → magma. Any rock can become any other.

⚑ PRELIMS TRAPS

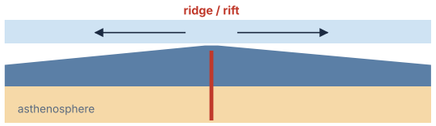
one Q a year

- S-waves never pass through liquid; P-waves pass through all states.
- The core is mainly **iron** (with nickel).
- Ocean floor is always **younger** (<200 Ma) than continental rock.
- Barren Island is **active**, in the Andaman Sea.
- Collision zones: quakes yes, volcanoes almost none.
- Hotspots sit **inside** plates.

④ PLATE BOUNDARIES — THE HERO DIAGRAM

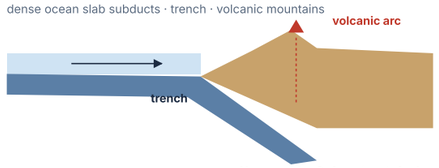
GS-I 2014 ×2, 2020 · Prelims 2024

DIVERGENT (constructive)
plates pull apart · new crust · shallow quakes



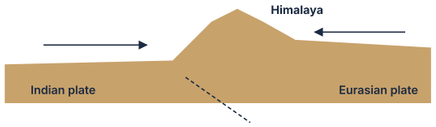
e.g. Mid-Atlantic Ridge, Iceland, East African Rift

CONVERGENT: ocean–continent
dense ocean slab subducts · trench · volcanic mountains



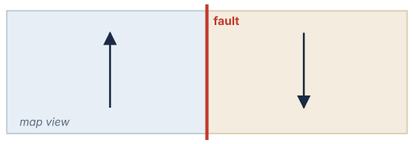
e.g. Nazca under S. America → Andes

CONVERGENT: continent–continent
buoyant crust collides · thickens · fold mountains, few volcanoes



e.g. India + Eurasia → Himalaya, Tibet

TRANSFORM (conservative)
plates slide past · no crust made or lost · shallow quakes



e.g. San Andreas Fault, California

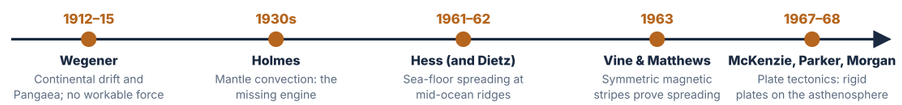
Boundary	Landform	Quakes / volcanoes	Example
Divergent	Ridge, rift valley	Shallow / fissure basalt	Mid-Atlantic Ridge, East African Rift
Ocean–continent	Trench + coastal fold range	Deep / explosive	Nazca → Andes
Ocean–ocean	Trench + island arc	Deep / arc volcanoes	Mariana, Japan, Indonesia
Continent–continent	Highest fold mountains	Strong / **almost none**	India + Eurasia → Himalaya
Transform	Fault, offset ridge	Shallow / none	San Andreas

Why fold mountains are at margins (2014): convergent boundaries lie along continental edges and collision sutures; margin sediments are squeezed. Old ranges (Appalachians, Urals, Aravallis) mark ancient sutures now inland.

Circum-Pacific (2020): 40,000 km horseshoe · trenches (Challenger Deep ~11 km) · island arcs · Andes, Rockies · **~81%** of largest quakes (Alpide ~17%, USGS) · most active volcanoes · deep-focus quakes · tsunamis · geothermal, copper.

③ DRIFT → SPREADING → PLATES

GS-I 2013 · Prelims 2025



Wegener's evidence: jigsaw fit (Bullard 1964) · same-age rocks Brazil–W Africa · tillite (Gondwana sediments in 6 landmasses) · Ghana gold from Brazil · fossils (Mesosaurus, Glossopteris, lemurs). **Failed:** pole-fleeing + tidal forces too weak.

Hess's evidence: rocks youngest at ridges · ocean floor <200 Ma vs continents ~3,200 Ma · thin sediments near ridges · symmetric magnetic stripes · shallow quakes at ridges, deep at trenches. **Plates move by** convection, ridge push, slab pull.

Intraplate quakes — Latur 1993: shocks away from any boundary.

Hotspot volcanoes — Hawaii, Réunion: volcanism in mid-plate.

Collision without volcanoes — Himalaya: continental crust will not subduct and melt.

Old mountains inland — Aravallis, Urals, Appalachians: ancient sutures, now worn down.

⑤ ISLAND ARCS · INDONESIA & PHILIPPINES

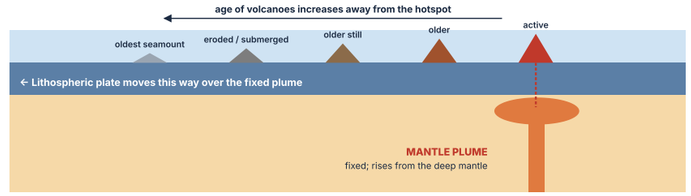
GS-I 2014 · 10M

- **Subduction arcs:** Indo-Australian under Sunda along the Java Trench → Sumatra–Java–Bali; Philippines squeezed from both sides (Philippine and Manila trenches).
- **Drowned Sunda Shelf:** post-glacial sea-level rise split Borneo, Sumatra, Java.
- **Coral** islands · **accreted** fragments (Sulawesi, Banda arc).

India: the same arc runs north as Andaman–Nicobar → Barren Island (only active volcano; erupted after 1991), the 2004 tsunami, Great Nicobar next to Sumatra.

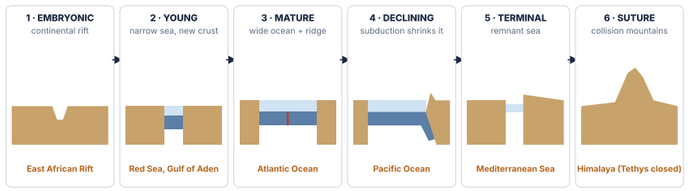
⑥ PLUMES & THE WILSON CYCLE

GS-I 2018 · 10M · 2025 · 15M



Pacific: Hawaii–Emperor chain — its bend records a change in the Pacific plate's direction.
India: Réunion plume → Deccan Traps; Kerguelen plume → Rajmahal Traps and the Ninetyeast Ridge.

Plume = narrow hot upwelling from the deep mantle; **hotspot** = its surface volcanism, usually **inside** a plate. Role: records plate motion (Hawaii–Emperor bend) · starts rifts (Afar) · builds large igneous provinces (**Réunion → Deccan ~66 Ma; Kerguelen → Rajmahal**) · minor plume push. Deep-plume origin still debated.



Opening (stages 1–3) → closing (4–6). The cycle can repeat: Pangaea formed and broke this way.

2025 answer frame: Wilson cycle + accretion + supercontinent cycle (Rodinia, Pangaea) + sea-level change + isostasy; India = Gondwana → Tethys → Himalaya.

THE VERDICT TO WRITE — The lithosphere is a mosaic of rigid plates riding on a weak asthenosphere, and nearly every fold mountain, trench, island arc, great earthquake and volcano sits on a plate boundary. **Wegener supplied the evidence, Hess the mechanism, plate tectonics the framework**; hotspots explain the exceptions; and the Wilson cycle shows continents and oceans are temporary arrangements — India's own journey from Gondwana to the Himalaya is the proof.